

1. What will be the output of the following code? Explain which namespace Python uses for x.

```
x = 10

def test():
    x = 20
    print(x)

test()
print(x)
```

2. What will be the output? Explain the **LEGB rule** step by step.

```
x = "Global"

def outer():
    x = "Enclosing"

    def inner():
        print(x)

    inner()

outer()
```

3. Identify the **Local**, **Enclosing**, **Global**, and **Built-in** names in the following program:

```
x = 10

def outer():
    y = 20

    def inner():
        z = 30
        print(x, y, z, len("Python"))

    inner()

outer()
```

4. What will be the output? Is x being shadowed?

```
x = 100

def outer():
    x = 200

    def inner():
        x = 300
        print(x)

    inner()
    print(x)

outer()
print(x)
```

5. What will happen when this program runs? Explain why.

```
count = 10

def update():
    count = count + 1
    print(count)

update()
```

6. Correct the following program using the global keyword so that the final output is 20.

```
x = 10

def update():
    x = 20

update()
print(x)
```

7. What will be the output? Explain the difference between the Local and Enclosing Namespace.

```
def outer():  
    x = 10  
  
    def inner():  
        x = 20  
        print(x)  
  
    inner()  
    print(x)  
  
outer()
```

8. What will be the output? Why does nonlocal change the value of x?

```
def outer():  
    x = 10  
  
    def inner():  
        nonlocal x  
        x = 50  
  
    inner()  
    print(x)  
  
outer()
```

9. Find the error and explain which namespace Python searches before raising the error.

```
def outer():  
    def inner():  
        print(name)  
  
    inner()  
  
outer()
```